

Week 1 Project: Electrical Lines and Basic Calculations

Week 1 Technical Report

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1 Introduction

This report details the work completed during the first week of the "Electrical Lines and Distribution Networks" project. The primary objectives for this initial phase were to establish a foundational understanding of line classification, analyze the fundamental properties of conductors and insulators, and perform basic electrical calculations for representative scenarios.

The methodologies employed adhere to the project requirements, utilizing MATLAB for data management, calculation, and visualization (with the graphs and charts later coded directly into LaTeX for higher quality). This document serves as the formal technical summary of the findings, with analysis grounded in Spanish electrical regulations, particularly the *Reglamento Electrotécnico para Baja Tensión* (REBT), and standard engineering principles. All calculations and data visualizations presented within this document are derived from the suite of MATLAB scripts developed for this project.

2 Problem 1: Line Classification and Selection Analysis

The initial task involved the classification of electrical lines for three distinct projects: an industrial facility, a residential complex, and a commercial center. A comprehensive database was established in MATLAB to manage these classifications. Structs were used for their convenience in grouping different data types into a single object, though tables could have also provided a clearer organization. (ref. `A1Line_Classification_And_Selection.m`).

2.1 Line Database and Classification

Lines were categorized based on three primary criteria: Voltage Level (Medium or Low Voltage), Application (Transmission, Distribution, Service or Interior), and Topology (Overhead, Underground, or Mixed). The voltage levels for the analyzed projects are visualized in Figure 1. As shown, the industrial and medium voltage commercial center lines operate at 20 kV, while the residential and low voltage commercial center lines operate at 400 V.

Table 1: Line Classification Database for Project Scenarios

Property	Industrial Installation	Residential Complex	Commercial Center (MV)	Commercial Center (LV)
Voltage (V)	20000	400	20000	400
Voltage Level	MV	LV	MV	LV
Application	Distribution	Distribution	Distribution	Service/Interior
Topology	Mixed	Underground	Overhead	Underground
Typical Conductor	XLPE Insulated Copper	PVC Insulated Copper	XLPE Insulated Aluminum	XLPE Insulated Copper

2.2 Conductor Material Distribution

An analysis of the typical conductor materials selected for these initial classifications reveals a strong preference for copper in low voltage and interior wiring, due to its longer lifespan and superior properties, such as a superior tolerance to heat and smaller cross section requirements. Aluminum is favored for the medium voltage feeder due to its lower cost, and its capacity to match copper's conductivity, with a lower overall weight, although with a bigger cross section. Its inferior strength properties can be compensated through

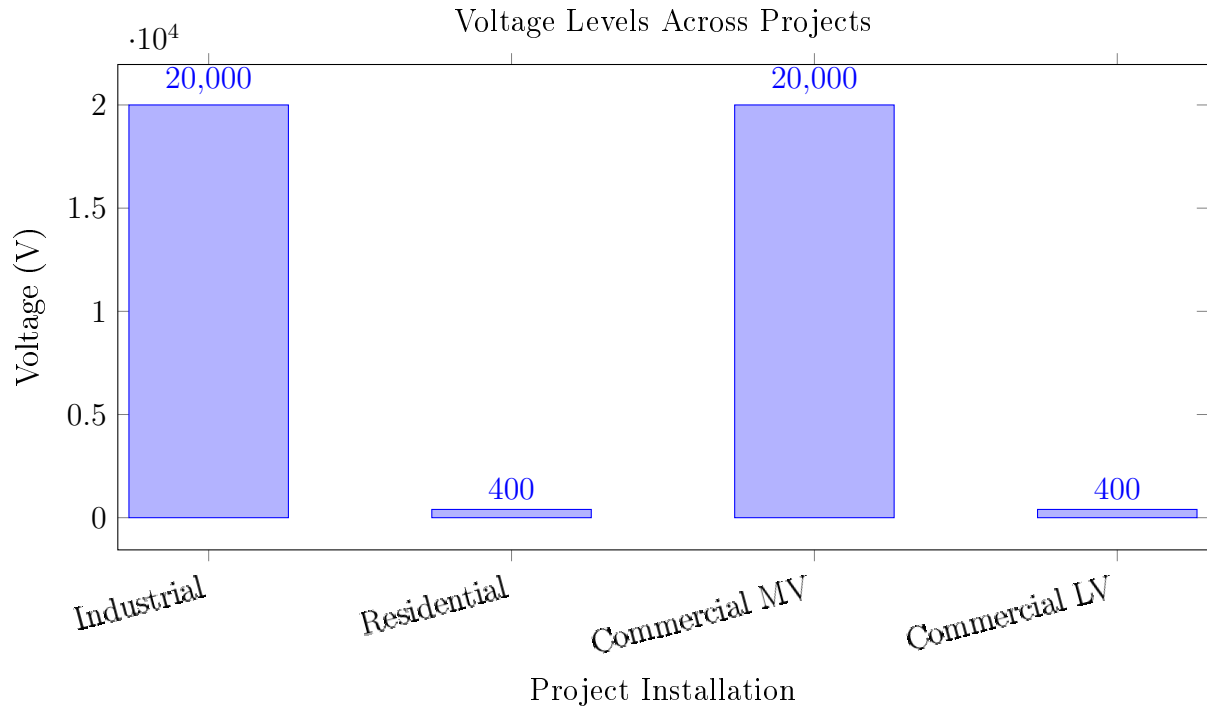


Figure 1: Comparison of nominal voltage levels for each project installation, based on data from The Project Guide

the use of steel (making an ACSR composite cable), although its susceptibility to warping due to heat or strain render its lifespan shorter. The relative distribution between the insulation type chosen for the different insulations is graphically shown in matlab with `A2 pie chart.m`, is presented in Figure 2. The selection reflects a common trade-off between cost (favoring aluminum for large distribution lines) and performance/regulatory requirements (favoring copper for service lines).

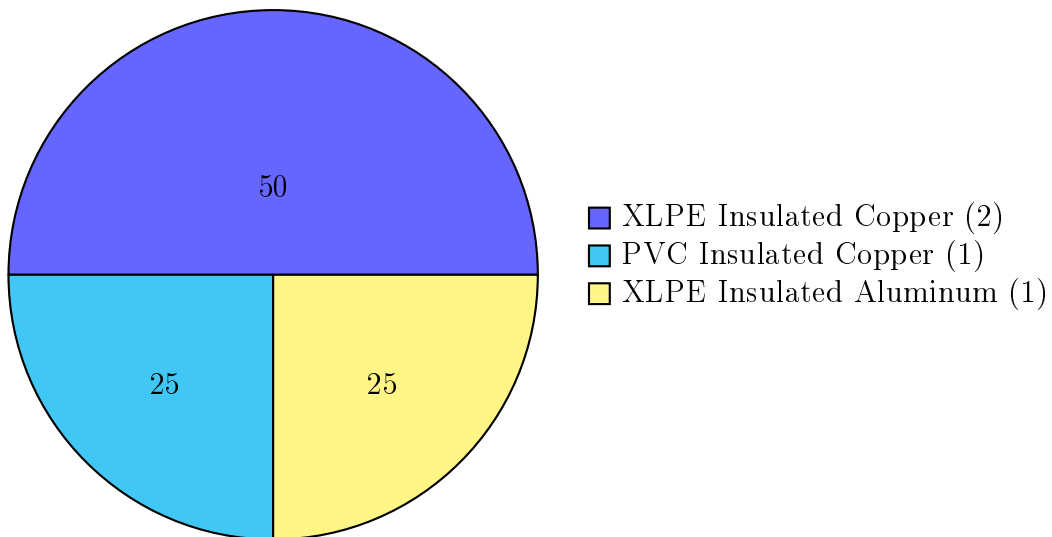


Figure 2: Distribution of conductor types across the initial project installations.

2.3 Interactive Selection Algorithm

To aid in future design decisions, an interactive selection algorithm was developed (ref. `A2LineTypeSelector.m`). This tool provides preliminary recommendations for line topology and conductor material based on user inputs for voltage, application location, and project lifetime. The underlying logic prioritizes overhead lines for main feeders due to cost and heat dissipation, while recommending underground lines for local distribution for safety and aesthetics. Material selection is guided by a longevity principle, favoring copper for projects exceeding a 30-year design life.

3 Problem 2: Conductor and Insulation Analysis

This section provides a detailed analysis of the mechanical and electrical properties of the primary conductor materials (Copper, Aluminum, and ACSR) and insulation systems (PVC and XLPE). The data for this analysis was compiled in the MATLAB scripts `B1_Conductor_Analysis.m` and `C3StrengthPropertyComparisons.m`.

3.1 Conductive Properties of the Conductors and Electrical Analysis

A key characteristic of any conductor is how well it conduces current. Copper is the most used conductor, for its relative cheapness when compared to other metals, and for its excellent conductive properties (being the base for the IACS system of measuring conductivity, in which Annealed Copper is 100% and Aluminum is just 61%). A table was made comparing the different situations the three students were faced with, each being given specific line length, power and power factor constraints, with which they were asked to calculate the overall line resistance, the line current, the voltage drop and the power loss. These calculations were done through the MATLAB Script `C2ElectricalConductivityWithLossV2`. The Maximum admissible current was also considered. In both cases, values were determined for a fixed set of parameters, which was sufficient for comparing the properties between materials. In the following graphs we visually compare the material's properties. We can appreciate how copper has the highest admissible current and the lowest voltage drop, whilst Aluminum and the ACSR composite have very similar conductive properties.

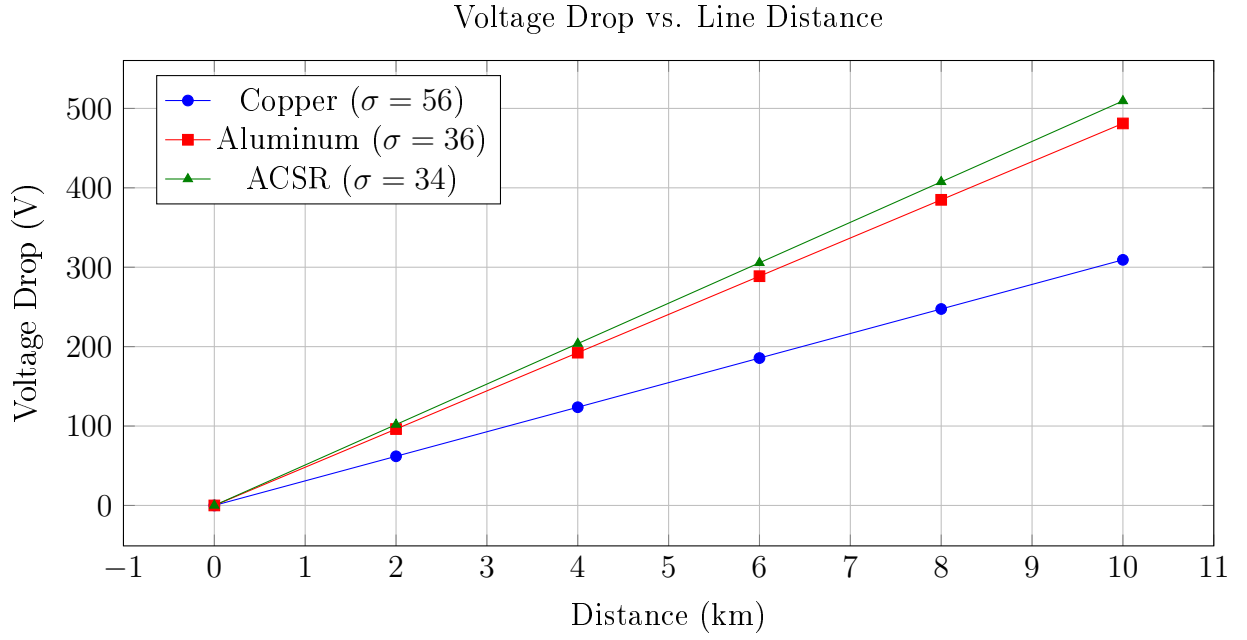


Figure 3: Progressive voltage drop for different conductor materials under a constant load.

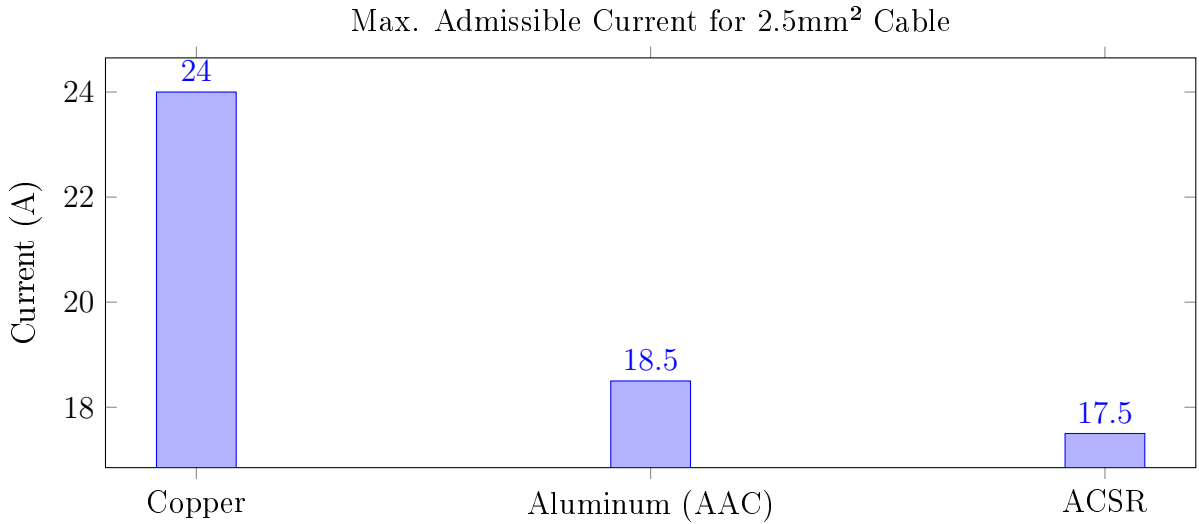


Figure 4: Comparison of maximum admissible current based on representative data for a 2.5mm² conductor.

3.2 Mechanical Properties of Conductors

The mechanical integrity of a conductor is critical, especially for overhead lines. The two key parameters analyzed were Tensile Strength and Young's Modulus.

Tensile Strength. This measures the maximum stress a material can withstand before breaking. As shown in Figure 5, ACSR (Aluminum-Steel Conductor Reinforced) exhibits vastly superior tensile strength due to its steel core. This makes it the ideal choice for long spans between towers in overhead lines. Copper is noticeably stronger than pure aluminum, but significantly weaker than the ACSR composite

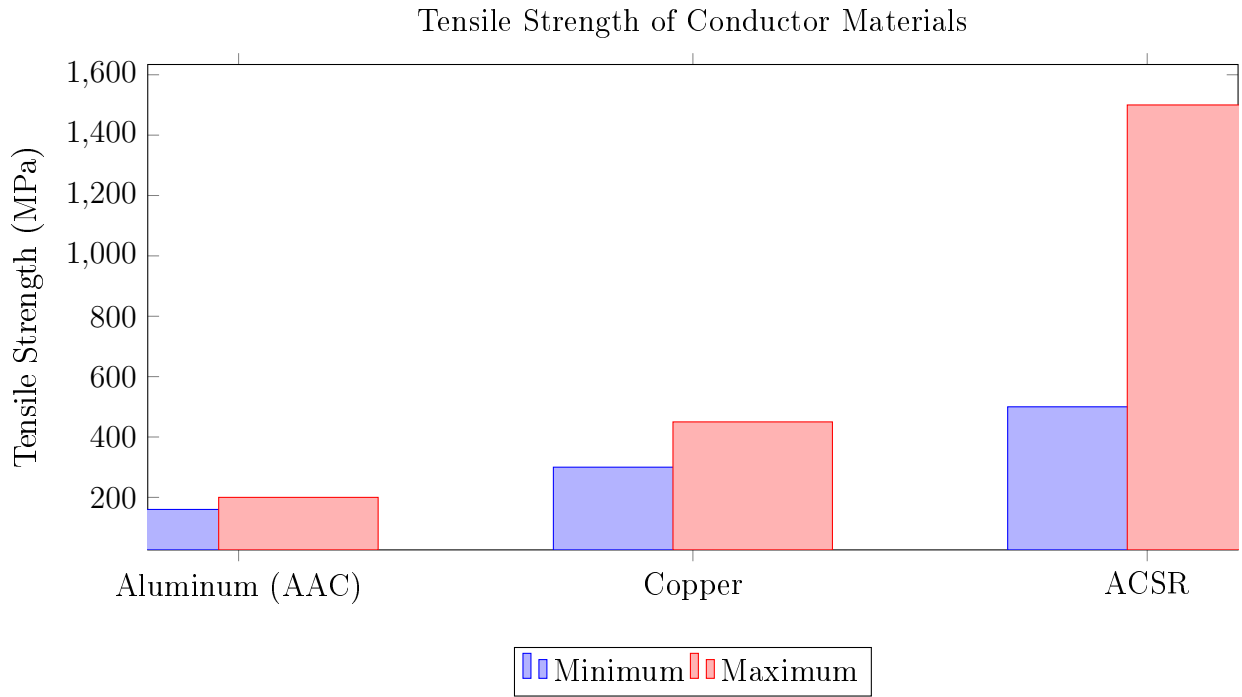


Figure 5: Comparison of minimum and maximum tensile strength ranges.

Young's Modulus. This value indicates the material's stiffness or resistance to elastic deformation under load. Figure 6 shows that copper is the stiffest material, which means it will sag less under its own weight for a given tension compared to aluminum. The ACSR conductor has a mixed stiffness, influenced by both its aluminum and steel components. It is important to note that the ranking the materials had when their Tensile Strength was being compared is not the same as their ranking upon considering Young's Modulus, the difference being that while copper may not be the strongest metal, it is the stiffest.

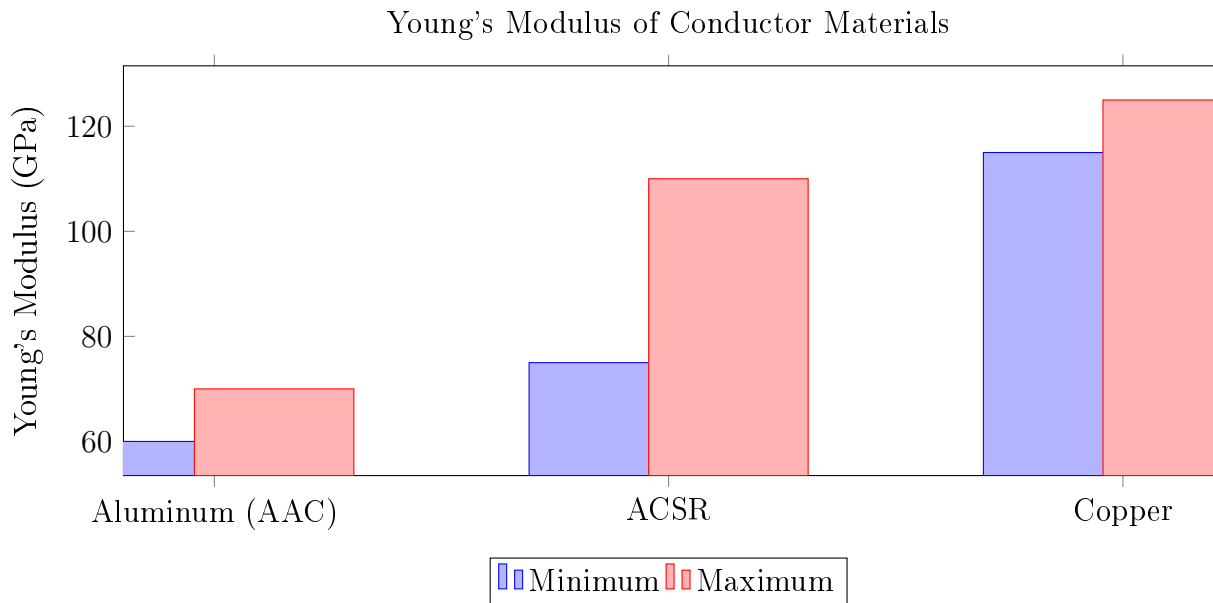


Figure 6: Comparison of Young's Modulus ranges, indicating material stiffness.

3.3 Insulation Systems Analysis

The choice of insulation is dictated by thermal performance, environmental conditions, and regulatory requirements. The analysis in `C1InsulationSystems.m` focused on PVC and XLPE. The most critical aspect to consider is the maximum continuous operating temperature: 70°C for PVC versus 90°C for XLPE.

PVC (Polyvinyl Chloride): A thermoplastic polymer. It is made flexible for cable use by adding plasticizers. It is a cost-effective and versatile insulator. It is characterized by a lower cost, high flexibility, and good resistance to oils and chemicals. However, it has a lower temperature rating, and produces dense, corrosive smoke (HCl gas) when burned. It also becomes brittle at low temperatures.

XLPE (Cross-linked Polyethylene) is a thermosetting polymer. The polyethylene molecules are permanently bonded ("cross-linked"), preventing them from melting or separating at higher temperatures, giving it superior thermal and mechanical stability. It has a higher operating temperature of 90°C, giving it its higher current rating for the same conductor size as PVC, as well as a greater mechanical toughness and water resistance. It does however have a higher material cost, and it is less flexible than PVC which can make installing it in tight spaces more difficult.

This thermal advantage allows an XLPE-insulated cable to carry significantly more current, and therefore more power, than a PVC-insulated cable of the same conductor size. This relationship is illustrated in Figure 7, which shows the calculated maximum power capacity for a standard 2.5 mm² copper conductor. This makes XLPE the mandatory choice for the high-power circuits where PVC does not meet the requirements as specified in REBT ITC-BT-07.

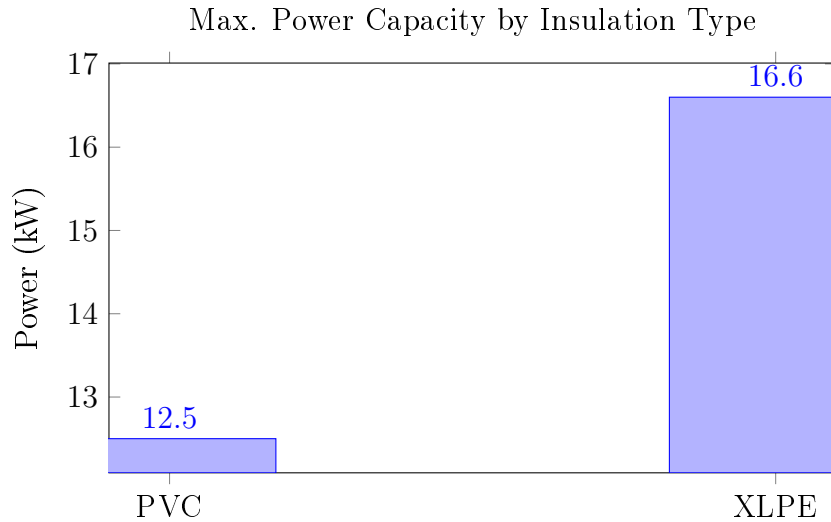


Figure 7: Calculated maximum three-phase power capacity at 400V for a 2.5 mm² copper conductor, limited by insulation temperature.

The wear of the conductor insulation has been modeled using a generic decay function that simplifies the Arrhenius equation used for modeling the decay of many systems. The wear of insulation material is due to heat which is treated as a function of temperature. The function used in the analysis is $R = R_0 - R_0 \cdot e^{-\alpha(T)}$, where R is the insulation resistance at a given temperature T , R_0 is the nominal resistance at a reference temperature, and the relative insulation resistance shows how much the material's thermal resistance

capacity has dropped at a certain temperature in comparison to the initial thermal resistance value at the reference temperature T_{ref} (e.g., 20°C), and α is the material-specific decay coefficient. This coefficient is the critical factor: PVC has a higher decay coefficient ($\alpha = 0.045$) than XLPE ($\alpha = 0.035$) (coefficients assumed for comparative purposes). This means that as temperature rises, PVC's insulation resistance drops more rapidly than XLPE's, indicating faster wear and degradation under thermal stress.

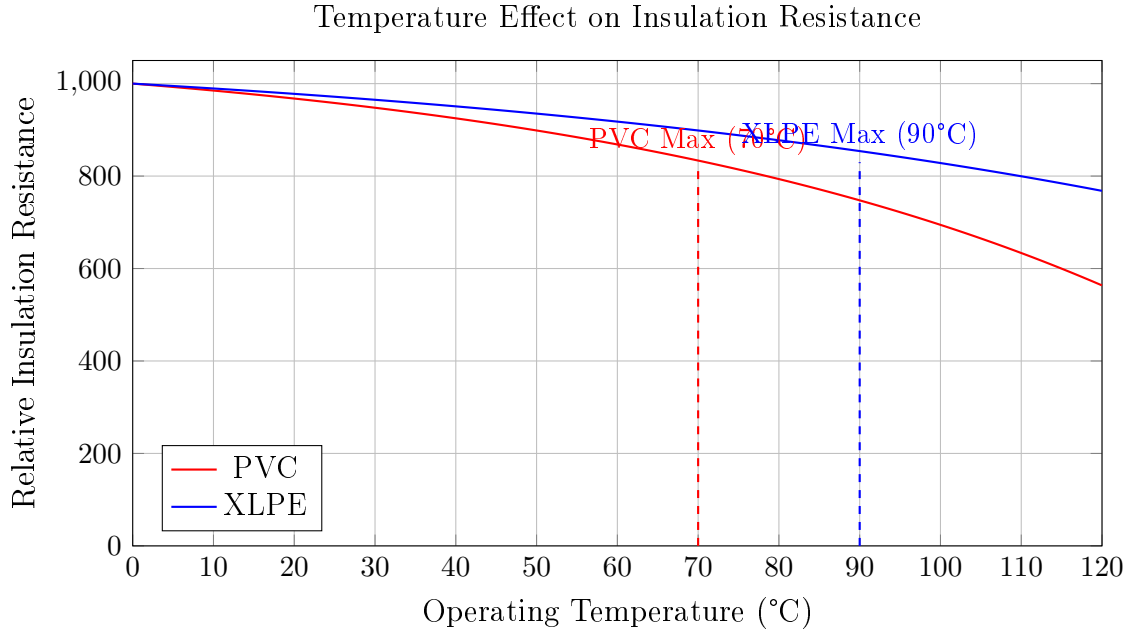


Figure 8: Comparison of insulation resistance decay for PVC and XLPE (More Curved Decay Model).

3.4 Economic Comparison

The economic comparison involves a trade-off between initial capital cost and long-term operating cost. Based on the assumed data, Aluminum is the cheapest conductor (700 €/km), followed by ACSR (850 €/km), and finally Copper (1200 €/km). For insulation, PVC is noted as having a lower initial cost than XLPE. Therefore, a cable using an Aluminum conductor with PVC insulation would typically have the lowest initial purchase price. However, operating costs are driven by power losses (I^2R), which are inversely related to conductivity. Copper has the highest conductivity ($\sigma = 56$), resulting in the lowest resistance and thus the lowest operating costs over the cable's lifetime. This is where the insulation choice becomes critical: XLPE, despite its higher initial cost, can safely operate at a higher temperature (90°C vs 70°C for PVC). This higher thermal rating allows the conductor to carry more current, which can enable the use of a smaller (and cheaper) conductor for the same power delivery. All of these trade offs lead to a concept known as the "economic section."

These charts illustrate the trade-off between initial cost and operating cost for different conductor materials, based on the assumed data. The initial conductor cost values per kilometer are taken from table B2 in the Appendix, using a reference of 35mm² cable of each type. The estimated energy loss cost over 30 years is estimated with the following formulas: Power loss per phase is $P_{loss} = I^2R$. For a balanced 3-phase system, total power loss is $P_{total} = 3 \times I^2R$. Total energy loss over a period is $E_{loss} = P_{total} \times \text{time} = 3 \times I^2R \times \text{time}$. R is the resistance per km at the assumed operating power. The final cost is based on an assumed cost of EUR 0.15 per kWh and a 30 year time span.

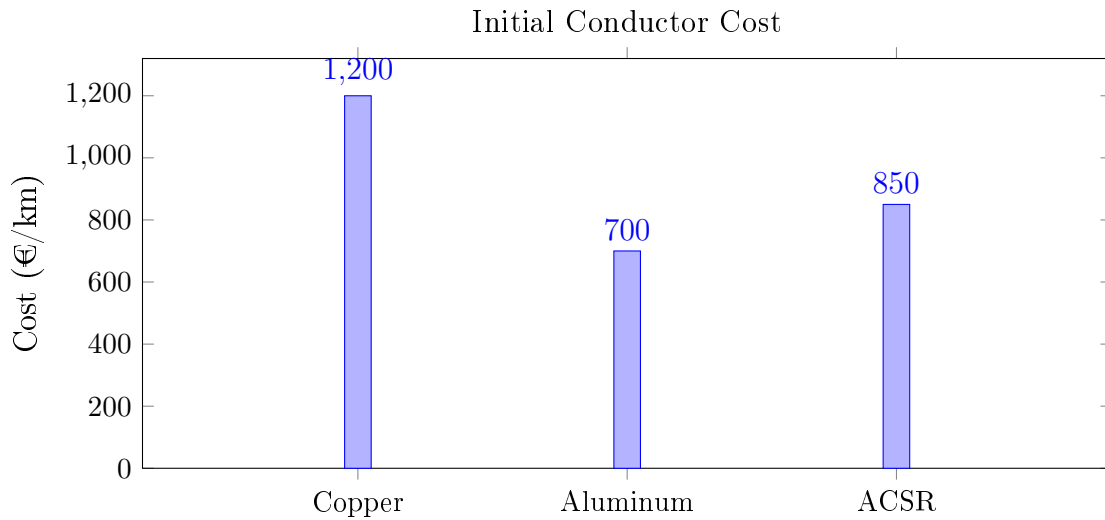


Figure 9: Initial material cost per kilometer.

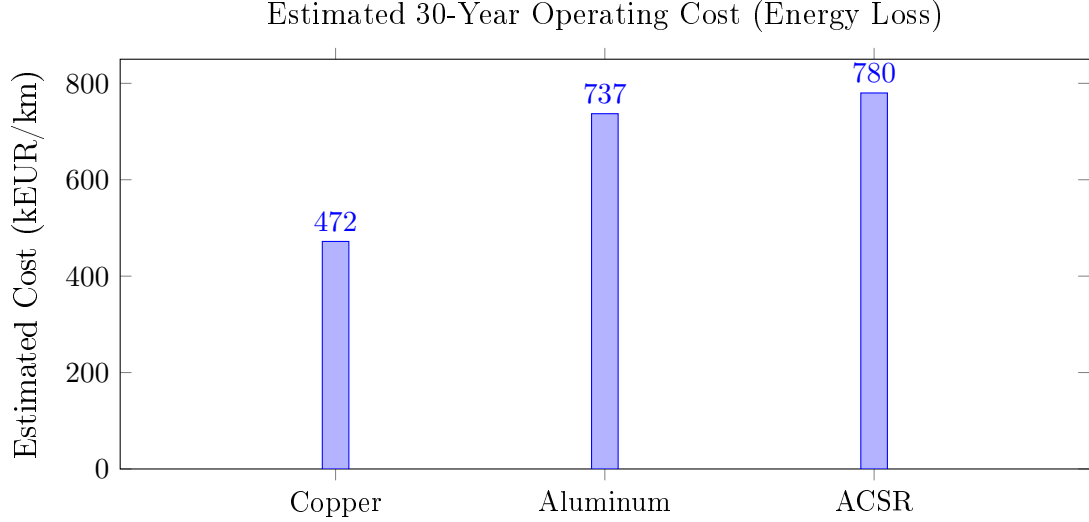


Figure 10: Estimated 30-year operating cost due to energy losses, based on assumed load (100A), area (50mm²), temp (50°C), and price (€0.15/kWh). For comparative purposes only.¹

4 Problem 3: Basic Line Calculations

To quantify the performance of a line, a set of fundamental calculations were implemented in the MATLAB script `D1BasicLineCalculations.m`. These calculations determine resistance, current, and voltage drop for the scenarios assigned to each student.

4.1 Fundamental Formulas

The following standard formulas were used for the analysis.

Resistance. The DC resistance (R) of a conductor is calculated based on its length (L), conductivity (σ), and cross-sectional area (s):

$$R = \frac{L}{\sigma \cdot s}$$

Three-Phase Current. The line current (I) in a balanced three-phase system is determined by the active power (P), line-to-line voltage (V_L), and the power factor ($\cos \varphi$):

$$I = \frac{P}{V_L \cdot \sqrt{3} \cdot \cos \varphi}$$

Voltage Drop. A simplified estimation for the per-phase voltage drop (ΔU_{phase}) in a BT line is given by the product of the line current and the phase resistance:

$$\Delta U_{\text{phase}} \approx I \cdot R$$

4.2 Scenario Analysis Results

The implemented functions were used to analyze the three student scenarios, with results compiled into a comparative table. The analysis assumes a standard 95 mm² copper conductor for consistent comparison.

Table 2: Comparative Results of Basic Line Calculations.

Parameter	Student A	Student B	Student C
Length (m)	500	1200	2500
Power (kW)	50	150	300
Power Factor	0.80	0.85	0.90
<i>Calculated Values</i>			
Resistance (Ω)	0.093	0.224	0.466
Current (A)	89.4	240.6	508.9
Voltage Drop (V/phase)	8.3	53.9	237.1

The results in Table 2 clearly demonstrate the impact of line length and transmitted power on electrical parameters. The scenario for Student C, with the longest line and highest power, experiences a severe voltage drop. This drop is far beyond the limits stipulated in the REBT (typically 3-6.5%, depending on the specific type of line), indicating that a 95 mm² conductor is critically undersized for this application. This initial calculation underscores the necessity of proper conductor dimensioning, which will be the focus of subsequent project weeks.

5 Conclusion

The first week of the project has successfully established the foundational framework for the design and analysis of electrical distribution networks. A structured database for line classification has been created, and the core mechanical and thermal properties of essential conductors and insulators have been analyzed and visualized.

The implementation of basic calculation formulas in MATLAB has provided critical insights into the performance of electrical lines under different load and length conditions. The results highlight the direct relationship between a line's physical characteristics and its electrical performance, particularly concerning voltage drop. It is evident from the initial analysis that compliance with REBT regulations is a primary constraint that will drive the design and dimensioning process in the upcoming weeks.

APPENDIX A

A References

1. **Reglamento Electrotécnico para Baja Tensión (REBT)**, Real Decreto 842/2002. Ministerio de Ciencia y Tecnología, Gobierno de España. Specifically consulted:
 - **ITC-BT-06**: Redes aéreas para distribución en baja tensión.
 - **ITC-BT-07**: Redes subterráneas para distribución en baja tensión.
 - **ITC-BT-19**: Instalaciones interiores o receptoras. Prescripciones generales y cálculo de intensidades máximas admisibles.
 - **ITC-BT-28**: Locales de Publica Concurrencia.
2. **Prysmian Group**, Technical Datasheet for *RZ1-K (AS) 0.6/1 kV* cables (Copper and Aluminum) and *ACSR Bare Overhead Conductor*. Representative data for conductors.
3. **General Cable (Prysmian Group)**, Technical Datasheet for *Hard-Drawn Bare Copper Conductor*. Representative data for bare copper conductors.
4. **ELEK Software**. (2023). *Estimating Cable Life Expectancy*. Retrieved October 18, 2025, from <https://elek.com/articles/estimating-cable-life-expectancy/>. This article provides context for the Arrhenius model application.

APPENDIX B

B Tables and Assumed Values

B.1 List of Tables

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B.2 Summary of Assumed Properties

The following table consolidates the key material properties and economic values assumed for the calculations and charts throughout this report. These values are representative and based on the references in Appendix A.

Table 3: Consolidated Material Properties and Assumed Values

Parameter	Copper	Aluminum	ACSR
<i>Electrical Properties</i>			
Conductivity (σ) [$\text{m}/(\Omega \cdot \text{mm}^2)$]	56	36	34
Max. Current for 2.5mm^2 (A)	24.0	18.5	17.5
<i>Mechanical Properties</i>			
Tensile Strength (MPa)	300 - 450	160 - 200	500 - 1500
Young's Modulus (GPa)	115 - 125	60 - 70	75 - 110
<i>Economic Properties</i>			
Cost ($\text{€}/\text{km}$ for 35mm^2)	6,250	1,650	2,100
<i>Assumed Values for Calculations</i>			
Standardized Conductor Width	95 mm^2 (for comparative calcs)		

APPENDIX C

C List of Formulas

C.1 Total Line Resistance

The electrical resistance of a conductor is calculated based on its material properties and physical dimensions:

$$R = \frac{L}{\sigma \cdot s} \quad (1)$$

C.2 Voltage Drop (Per Phase)

For this preliminary analysis, the voltage drop per phase is estimated using Ohm's Law:

$$\Delta U_{\text{phase}} = I \cdot R \quad (2)$$

C.3 Three-Phase Power Loss

The total power lost as heat in a balanced three-phase line is given by:

$$P_{\text{loss}} = 3 \cdot I^2 \cdot R \quad (3)$$

C.4 Insulation Aging Estimation

The Arrhenius model can be simplified to estimate the degradation of an insulation's resistance over time due to temperature. This is crucial for predicting the useful lifespan of a cable.

$$R_{\text{ins}} = R_0 e^{-\alpha(T-T_0)} \quad (4)$$

Where:

- R_{ins} is the insulation resistance at the operating temperature.

- R_0 is the initial insulation resistance at a reference temperature.
- α is the material's temperature coefficient for insulation degradation.
- T is the operating temperature in degrees Celsius ($^{\circ}\text{C}$).
- T_0 is the reference temperature (e.g., 20°C).

APPENDIX D

D MATLAB Source Code

D.1 AALine _Classification _And _Selection.m

```

1 % Create a table to store line classifications
2 % Using a structure to organize line data
3 powerLines= struct();
4
5 % Student A
6 powerLines.IndustrialFacility.Proyect = 'Industrial Instalation';
7 powerLines.IndustrialFacility.Voltage = 20000; % Volts
8 powerLines.IndustrialFacility.VoltageLevel = 'MV';
9 powerLines.IndustrialFacility.Application = 'Distribution';
10 powerLines.IndustrialFacility.Topology = 'Mixed';
11 powerLines.IndustrialFacility.TypicalConductor = 'XLPE Insulated Copper'
12     ;
13 %Student B
14 powerLines.ResidentialComplex.Proyect = 'Residential Complex';
15 powerLines.ResidentialComplex.Voltage = 400; % Volts
16 powerLines.ResidentialComplex.VoltageLevel = 'LV';
17 powerLines.ResidentialComplex.Application = 'Distribution';
18 powerLines.ResidentialComplex.Topology = 'Underground';
19 powerLines.ResidentialComplex.TypicalConductor = 'PVC Insulated Copper';
20
21 %Student C
22     %Linea de Media Tension
23 powerLines.CommercialCentreMV.Proyect = 'Commercial Centre (MV)';
24 powerLines.CommercialCentreMV.Voltage = 20000; % Volts
25 powerLines.CommercialCentreMV.VoltageLevel = 'MV';
26 powerLines.CommercialCentreMV.Application = 'Distribution';
27 powerLines.CommercialCentreMV.Topology = 'Overhead';
28 powerLines.CommercialCentreMV.TypicalConductor = 'XLPE Insulated
29     Aluminum';
30
31     %Linea de Baja Tension
32 powerLines.CommercialCentreLV.Proyect = 'Commercial Centre (LV)';
33 powerLines.CommercialCentreLV.Voltage = 400; % Volts
34 powerLines.CommercialCentreLV.VoltageLevel = 'LV';
35 powerLines.CommercialCentreLV.Application = 'Service/Interior';
36 powerLines.CommercialCentreLV.Topology = 'Underground';
37 powerLines.CommercialCentreLV.TypicalConductor = 'XLPE Insulated Copper'
38     ;
39 % Display the database in the command window
40 disp('Commercial Center Line Database:');
```

```

40 disp(powerLines.IndustrialFacility)
41 disp(powerLines.ResidentialComplex)
42 disp(powerLines.CommercialCentreMV)
43 disp(powerLines.CommercialCentreLV)
44
45 %
46 % Generate a visual comparison chart
47 figure; % Create a new window for the plot
48 voltages = [powerLines.IndustrialFacility.Voltage, powerLines.
49             ResidentialComplex.Voltage, powerLines.CommercialCentreMV.Voltage,
50             powerLines.CommercialCentreLV.Voltage];
51 labels = {'Industrial Facility', 'Residential Complex', 'Commercial
52           Centre (Medium Voltage)', 'Commercial Centre (Low Voltage)'};
53 bar(voltages); % Create a bar chart
54 set(gca, 'xticklabel', labels);
55 title('Voltage Levels in the Different Electrical Instalacion Projects')
56 ;
57 ylabel('Voltage (V)');

```

D.2 ABLineTypeSelector.m

```

1 % Interactive Electrical Line Designer
2 % Project: Electrical Lines and Distribution Networks
3 % Task: Interactively guide a user to select line topology and conductor
4     material.
5
6 % --- Initialization ---
7
8 clear; clc; close all;
9
10 fprintf('-----\n');
11 fprintf('    Interactive Electrical Line Design Assistant    \n');
12 fprintf('-----\n\n');
13
14 % --- User Input ---
15 fprintf('Please provide the following project parameters:\n');
16
17 voltage_V = input('1. Enter the nominal voltage in Volts (e.g., 400 or
18                 20000): ');
19
20 fprintf('2. Select the line application type:\n');
21 fprintf('    [1] Main Feeder (Primary distribution line)\n');
22 fprintf('    [2] Local Distribution (Secondary network, service drops)\n'
23 );
24 location_choice = input('    Enter your choice [1 or 2]: ');
25 if location_choice == 1
26     location = 'main-feeder';
27 else
28     location = 'local-distribution';
29 end
30 lifetime_years = input('3. Enter the expected project lifetime in years
31                       (e.g., 25): ');
32
33 % --- Analysis and Recommendation Engine ---
34 lineTopology = selectLineTopology(voltage_V, location);
35 conductorMaterial = selectConductorMaterial(lifetime_years);
36

```

```

31 % --- Display Results ---
32 fprintf('\n-----\n');
33 fprintf('          PRELIMINARY DESIGN RECOMMENDATION          \n');
34 fprintf('-----\n');
35 fprintf('Based on your inputs:\n');
36 fprintf(' * Voltage: %.0f V\n', voltage_V);
37 fprintf(' * Application: %s\n', strrep(location, '_', ' '));
38 fprintf(' * Lifetime: %.0f years\n\n', lifetime_years);
39 fprintf('Recommendations:\n');
40 fprintf(' * %s\n', conductorMaterial);
41 fprintf(' * %s\n', lineTopology);
42 fprintf('-----\n\n');
43 fprintf('Disclaimer: This is a preliminary recommendation. A detailed\n\n');
44     engineering study is required.\n');
45
46 % --- Helper Functions ---
47 function [lineType] = selectLineTopology(voltage, location)
48     if voltage > 1000 && strcmpi(location, 'main_feeder')
49         lineType = 'Topology: Overhead MT Line Recommended for cost-
50 effectiveness and heat dissipation.';
51     elseif voltage <= 1000 && strcmpi(location, 'local_distribution')
52         lineType = 'Topology: Underground BT Line Recommended for safety
53 and aesthetics in populated areas.';
54     else
55         lineType = 'Topology: Mixed/Special Case - Requires detailed
56 analysis (e.g., Underground MT).';
57     end
58 end
59
60 function [material] = selectConductorMaterial(years)
61     if years > 30
62         material = 'Conductor Material: Copper Recommended for its high
63 durability and lower long-term losses.';
64     else
65         material = 'Conductor Material: Aluminum Recommended as a cost-
66 effective solution for shorter-term projects.';
67     end
68 end

```

D.3 ABpie chart.m

```

1 % MATLAB Script for Week 1 Project: Conductor Type Visualization
2 % Task: Generate a pie chart of conductor types for different
3     installations.
4
5 % --- Configuration ---
6 clear; clc; close all;
7
8 % --- Data Definition ---
9 powerLines.IndustrialFacility.TypicalConductor = 'XLPE Insulated Copper';
10 powerLines.ResidentialComplex.TypicalConductor = 'PVC Insulated Copper';
11 powerLines.CommercialCentreMV.TypicalConductor = 'XLPE Insulated
12     Aluminum';
13 powerLines.CommercialCentreLV.TypicalConductor = 'XLPE Insulated Copper';
14
15

```

```

13 % --- Data Processing ---
14 conductors = {
15     powerLines.IndustrialFacility.TypicalConductor;
16     powerLines.ResidentialComplex.TypicalConductor;
17     powerLines.CommercialCentreMV.TypicalConductor;
18     powerLines.CommercialCentreLV.TypicalConductor
19 };
20 conductorCategories = categorical(conductors);
21 uniqueConductors = categories(conductorCategories);
22 conductorCounts = countcats(conductorCategories);
23 percentages = 100 * conductorCounts / sum(conductorCounts);
24
25 % --- Visualization ---
26 fig = figure('Name', 'Conductor Type Distribution', 'NumberTitle', 'off'
27 );
28 set(fig, 'Color', 'white');
29 p = pie(conductorCounts);
30 title('Distribution of Conductor Types Across Installations', 'FontSize',
31     14, 'FontWeight', 'bold');
32
33 % --- Chart Enhancement ---
34 legendLabels = cell(size(uniqueConductors));
35 for i = 1:length(uniqueConductors)
36     legendLabels{i} = sprintf('%s (%d, %.1f%%)', uniqueConductors{i},
37         conductorCounts(i), percentages(i));
38 end
39 lgd = legend(legendLabels, 'Location', 'northeastoutside', 'FontSize',
40     10);
41 title(lgd, 'Conductor Key');

```

D.4 BAConductor_Analisis.m

```

1 % Create a structure for conductor properties
2 Conductors = struct();
3
4 %data for Copper
5 Conductors.Copper.name = 'Copper';
6 Conductors.Copper.conductivity = 56; % in m/(Ohm*mm^2)
7 Conductors.Copper.cost_per_km = 1200; % Example cost in Euros
8 Conductors.Copper.temp_coeff = 0.0039; % Alpha coefficient for
   resistance
9 Conductors.Copper.tensileStrenghtMax = 450;
10 Conductors.Copper.tensileStrenghtMin = 300;
11 Conductors.Copper.youngModuleMax = 125;
12 Conductors.Copper.youngModuleMin = 115;
13
14 %data for Aluminum
15 Conductors.Aluminum.name = 'Aluminum';
16 Conductors.Aluminum.conductivity = 36;
17 Conductors.Aluminum.cost_per_km = 700; % Example cost
18 Conductors.Aluminum.temp_coeff = 0.0040;
19 Conductors.Aluminum.tensileStrenghtMax = 160;
20 Conductors.Aluminum.tensileStrenghtMin = 200;
21 Conductors.Aluminum.youngModuleMax = 70;
22 Conductors.Aluminum.youngModuleMin = 60;
23
24 %data for Aluminum-Steel Composite

```

```

25 Conductors.ACSR.name = 'Aluminum-Steel (ACSR)';
26 Conductors.ACSR.conductivity = 34; % Effective conductivity is slightly
    lower
27 Conductors.ACSR.cost_per_km = 900; % Example cost
28 Conductors.ACSR.temp_coeff = 0.0040;
29 Conductors.ACSR.tensileStrenghtMax = 1500;
30 Conductors.ACSR.tensileStrenghtMin = 500;
31 Conductors.ACSR.youngModuleMax = 110;
32 Conductors.ACSR.youngModuleMin = 75;

```

D.5 CBElectricalConductivityWithPowerLossV2.m

```

1 % --- Electrical Performance Comparison of Conductor Materials ---
2
3 % --- Initial Setup ---
4 clear; clc; close all;
5
6 % --- Define Constants and Material Properties ---
7 S_va = 1000; V_v = 400; s_mm2 = 2.5;
8 rebt_v_drop_percent = 5; rebt_v_limit = (rebt_v_drop_percent / 100) *
    V_v;
9
10 materials = {'Copper', 'Aluminum (AAC)', 'ACSR'};
11 sigma = [56, 36, 34]; % m/(Ohm*mm^2)
12 i_max_amps = [24, 18.5, 17.5]; % A
13 cost_per_km = [6250, 1650, 2100]; % Euros
14
15 % --- Calculations ---
16 I_load_a = S_va / (V_v * sqrt(3));
17 L_m = (0:100:10000); % 0 to 10 km
18 R_ohm = (1 ./ sigma)' * (L_m / s_mm2);
19 V_drop_v = I_load_a * R_ohm;
20 Power_loss_w = (I_load_a^2) * R_ohm;
21 cost = cost_per_km' * (L_m / 1000);
22
23 % --- Visualization ---
24 figure('Position', [100 100 1200 800]);
25
26 % Subplot 1: Voltage Drop
27 subplot(2, 2, 1);
28 hold on;
29 for i = 1:length(materials)
30     plot(L_m/1000, V_drop_v(i,:), 'LineWidth', 2);
31 end
32 yline(rebt_v_limit, '--r', 'REBT Limit (20V)', 'LineWidth', 2);
33 hold off;
34 title('Progressive Voltage Drop vs. Distance');
35 xlabel('Distance (km)'); ylabel('Voltage Drop (V)');
36 legend(materials, 'Location', 'northwest'); grid on;
37
38 % Subplot 2: Power Loss
39 subplot(2, 2, 2);
40 hold on;
41 for i = 1:length(materials)
42     plot(L_m/1000, Power_loss_w(i,:), 'LineWidth', 2);
43 end
44 hold off;

```

```

45 title('Progressive Power Loss (I^2R) vs. Distance');
46 xlabel('Distance (km)'); ylabel('Power Loss (W)');
47 legend(materials, 'Location', 'northwest'); grid on;
48
49 % Subplot 3: Max Current Bar Chart
50 subplot(2, 2, 3);
51 bar(i_max_amps);
52 set(gca, 'xticklabel', materials);
53 title(sprintf('Max. Admissible Current for %.1f mm^2 Cable', s_mm2));
54 ylabel('Current (A)'); grid on;
55
56 % Subplot 4: Cost vs Distance
57 subplot(2, 2, 4);
58 hold on;
59 for i = 1:length(materials)
60     plot(L_m/1000, cost(i,:), 'LineWidth', 2);
61 end
62 hold off;
63 title('Progressive Installation Cost vs. Distance');
64 xlabel('Distance (km)'); ylabel('Total Cost (EUR)');
65 legend(materials, 'Location', 'northwest'); grid on;

```

D.6 CCStrenghtPropertyComparisons.m

```

1 % --- Mechanical Properties Data Definition ---
2 material_names = {'Aluminum (AAC)', 'Copper', 'ACSR'};
3
4 tensile_strength_mpa = [
5     160, 200; % Aluminum (AAC)
6     300, 450; % Copper
7     500, 1500 % ACSR
8 ];
9
10 youngs_modulus_gpa = [
11     60, 70; % Aluminum (AAC)
12     115, 125; % Copper
13     75, 110 % ACSR
14 ];
15
16 % --- Graph 1: Tensile Strength ---
17 min_strength = tensile_strength_mpa(:, 1);
18 [~, sort_order] = sort(min_strength);
19 sorted_names_ts = material_names(sort_order);
20 sorted_tensile_strength = tensile_strength_mpa(sort_order, :);
21
22 figure;
23 bar(sorted_tensile_strength);
24 title('Tensile Strength of Conductor Materials', 'FontSize', 14, '
    FontWeight', 'bold');
25 ylabel('Tensile Strength (MPa)', 'FontSize', 12);
26 xlabel('Conductor Material', 'FontSize', 12);
27 set(gca, 'xticklabel', sorted_names_ts);
28 legend({'Minimum Strength', 'Maximum Strength'}, 'Location', 'northwest'
    );
29 grid on;
30
31 % --- Graph 2: Young's Modulus ---

```

```

32 avg_modulus = mean(youngs_modulus_gpa, 2);
33 [~, sort_order_ym] = sort(avg_modulus);
34 sorted_names_ym = material_names(sort_order_ym);
35 sorted_youngs_modulus = youngs_modulus_gpa(sort_order_ym, :);
36
37 figure;
38 bar(sorted_youngs_modulus);
39 title("Young's Modulus of Conductor Materials", 'FontSize', 14, '
    FontWeight', 'bold');
40 ylabel("Young's Modulus (GPa)", 'FontSize', 12);
41 xlabel('Conductor Material', 'FontSize', 12);
42 set(gca, 'xticklabel', sorted_names_ym);
43 legend({'Minimum Modulus', 'Maximum Modulus'}, 'Location', 'northwest');
44 grid on;

```

D.7 CDInsulationSystems.m

```

1 % --- Comprehensive Analysis of PVC and XLPE Insulation Systems ---
2
3 % --- Initial Setup ---
4 clear; clc; close all;
5
6 % --- Define Constants and Insulation Properties ---
7 V_v = 400; s_mm2 = 2.5;
8 insulators = {'PVC', 'XLPE'}; max_temp_C = [70, 90]; i_max_insulation =
    [18, 24];
9
10 % --- Calculations ---
11 max_power_kw = (V_v * sqrt(3) * i_max_insulation) / 1000;
12
13 % --- Visualization: Bar Chart of Max Power Capacity ---
14 figure('Name', 'Power Capacity by Insulation Type');
15 b = bar(max_power_kw, 'FaceColor', 'flat');
16 b.CData(1,:) = [0.85 0.325 0.098]; b.CData(2,:) = [0 0.447 0.741];
17 title(sprintf('Max. Power Capacity of a %.1f mm^2 Cu Cable', s_mm2));
18 ylabel('Power (kW)'); set(gca, 'xticklabel', insulators); grid on;
19 text(1:length(max_power_kw), max_power_kw, num2str(max_power_kw, '%.1f
    kW'), 'vert', 'bottom', 'horiz', 'center');
20
21 % --- Detailed Report in Command Window ---
22 fprintf('--- INSULATION SYSTEMS ANALYSIS ---\n');
23 fprintf('PVC: Max Temp %dC. XLPE: Max Temp %dC.\n', max_temp_C(1),
    max_temp_C(2));

```

D.8 DDBasicLineCalculations.m

```

1 % --- Basic Line Calculations for Student Scenarios (Problem 3) ---
2
3 % --- Initial Setup ---
4 clear; clc; close all;
5
6 % --- 1. Define Calculation Functions ---
7 function R = calculateResistance(L, sigma, s)
8     R = L / (sigma * s);
9 end

```

```

10 function I = calculateCurrent3ph(P_kW, V_V, cos_phi)
11     I = (P_kW * 1000) / (V_V * sqrt(3) * cos_phi);
12 end
13 function deltaU = estimateVoltageDropPhase(I, R)
14     deltaU = I * R;
15 end
16
17 % --- 2. Define Common Parameters ---
18 V_line_V = 400;
19 s_conductor_mm2 = 95;
20 sigma_copper = 56;
21
22 % --- 3. Define Student-Specific Scenario Data ---
23 student_data = table(...
24     {'A'; 'B'; 'C'}, ...
25     [500; 1200; 2500], ...
26     [50; 150; 300], ...
27     [0.80; 0.90; 0.85], ...
28     'VariableNames', {'Student', 'Length_m', 'Power_kW', 'PowerFactor'}
29 );
30
31 % --- 4. Perform Calculations ---
32 calculated_resistance = zeros(height(student_data), 1);
33 calculated_current = zeros(height(student_data), 1);
34 calculated_vdrop_phase = zeros(height(student_data), 1);
35
36 for i = 1:height(student_data)
37     L = student_data.Length_m(i);
38     P = student_data.Power_kW(i);
39     pf = student_data.PowerFactor(i);
40
41     calculated_resistance(i) = calculateResistance(L, sigma_copper,
42         s_conductor_mm2);
43     calculated_current(i) = calculateCurrent3ph(P, V_line_V, pf);
44     calculated_vdrop_phase(i) = estimateVoltageDropPhase(
45         calculated_current(i), calculated_resistance(i));
46 end
47
48 % --- 5. Combine and Display Results ---
49 results_table = student_data;
50 results_table.Resistance_Ohms = calculated_resistance;
51 results_table.Current_Amps = calculated_current;
52 results_table.VoltageDrop_V_per_Phase = calculated_vdrop_phase;
53
54 disp('--- Comparative Analysis of Student Scenarios ---');
55 disp(results_table);

```